

Plant-Herbivore Coevolution in a Spatially and Genetically Explicit Model

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Abstract A coevolutionary model was developed to test interactions between diploid plants and herbivores using genetic algorithms on a spatial lattice. Simulated plants carried defensive genes and herbivores carried genes coding for resistance (metabolism of herbivore defense) in gene-for-gene synchrony. Collectively these genes are referred to as defensive/resistance genes (DR-genes). Genes were linked on chromosomes. Regulatory genes modified both dominance at these DR loci and the tradeoff cost involved in producing either defense or resistance. We tested the effects of varying a) the number of DR-loci, b) the ratio of the number of herbivore:plant generations, c) the shape (square vs. long and thin) and function (torus vs. island) of the lattice, and d) herbivore encounter rate on plant progeny dispersal distance.

Increasing both the number of DR-genes and the ratio of herbivore:plant generations caused a tighter coevolutionary response between plants and herbivores. Plant defense was highly sensitive to herbivory but not to increasing encounter rates. Plant DR-genes were selectively disadvantageous with only one locus but selectively favored with two or more loci. Increasing the number of herbivore:plant generations caused increased fluctuations in herbivore resistance gene frequencies and a decrease in the lag time in herbivore response to changes in plant defensive gene frequencies. The relationship between heterotroph and autotroph DR-genes increased exponentially with increasing numbers of DR-loci. This relationship suggests that autotrophs benefit from increased diversity of defense that causes a relative increase in cost for the heterotrophs. The shape of the lattice interacted with lattice function, resulting in high species persistence on wraparound habitats and the greatest extinction likelihood on rectangular islands. Low to moderate herbivore encounter rates increased plant progeny dispersal distance while high herbivore encounter rates tended to reduce dispersal distance. The frequencies of genes coding for plant defense and herbivore resistance were dynamic for thousands of generations, despite the homogeneous lattice. This interaction may increase extinction probabilities in fragmented habitats.

Keywords

artificial life, evolution, coevolution, plant-herbivore interactions, genetic algorithms

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I Introduction

Coevolution between species has received a great deal of empirical and theoretical attention over the past several decades. The coevolution of phenotypes has great significance to humans, particularly with respect to human pathogens [1] and agricultural systems [36]. Coevolution, unfortunately, is a term that has been used to describe many types of species interactions. Coevolution should be considered a particular type of interaction that influences gene frequencies between interacting species [21]. The interaction between species may be weak, with one species tracking genetic changes in another without reciprocal effects. We consider that reciprocal effects constitute coevolution. Under such circumstances changes in one species affect gene frequencies of a second species, which, in turn, affect gene frequencies in the first species.

An example of reciprocal coevolution can be found in gene-for-gene systems, where genes are directly mapped between coevolving species [9, 33]. Flor [9] identified gene-for-gene interactions in a host-parasite system, identifying 25 genes coding for resistance in flax (*Linum usitatissimum* L.), matched gene-for-gene in a flax rust (*Melampsora lini*) (Pers.) Lev. Thompson and Burdon [33] argue that host-pathogen gene-for-gene systems are likely to be widespread [but see 15, 37]. Despite debate on the abundance of gene-for-gene systems, there is general agreement that coevolutionary interactions occur between plants and herbivores for defense and resistance, respectively [11].

In addition, coevolution may be diffuse where interactions are likely to be influenced by more than one species. For instance, Schmitz [30] tested the effect of removing spider predators on herbivorous insect populations. He found that the removal of spiders increased the biomass of resistant plants by increasing herbivore biomass, suggesting microevolution in plant resistant gene frequencies. The coevolutionary relationship between plants and herbivores may be particularly important in spatially structured plant-herbivore systems where biological control results in variable encounter rates [25].

The coevolutionary relationship between plants and herbivores depends on associated costs in the production of resistance or detoxifying factors [31, 32]. The presence of costs for producing defensive cyanogenic glycosides in *Trifolium*, for instance, has been well demonstrated [5]. Associated costs with herbivore tolerance and detoxification, on the other hand, is less certain. The evolution of the cost for insecticide resistance (tradeoff) has been documented for the Australian sheep blowfly (*Lucilia cuprina*) [29, reviewed by 31]. Blowflies carrying a resistance allele exhibited reduced fitness only when introduced into a pesticide-susceptible population. McKenzie et al. [29] concluded that after successive generations resistant genotypes evolved reduced reproductive cost for the resistance factor, suggesting evolution in the tradeoff associated with pesticide resistance. Therefore, we allowed costs associated with resistance/detoxifying factors (tradeoffs) to evolve by altering the intensity of plant-herbivore interactions.

2 Modeling Space in Coevolutionary Systems

Spatially explicit, stochastic models using cellular automata [8] and interacting particle systems [7] are now widely used to test the importance of space in ecological systems. These techniques, borrowed from mathematics and physics, have increased our understanding of spatially dependent ecologically based population processes. These approaches involve either analytical treatment [6] or a more open form of simulation [2, 13, 35], as is used in this coevolutionary model. Cellular automata models are typified by cells with transitory, discrete states that interact within a local neighborhood. They

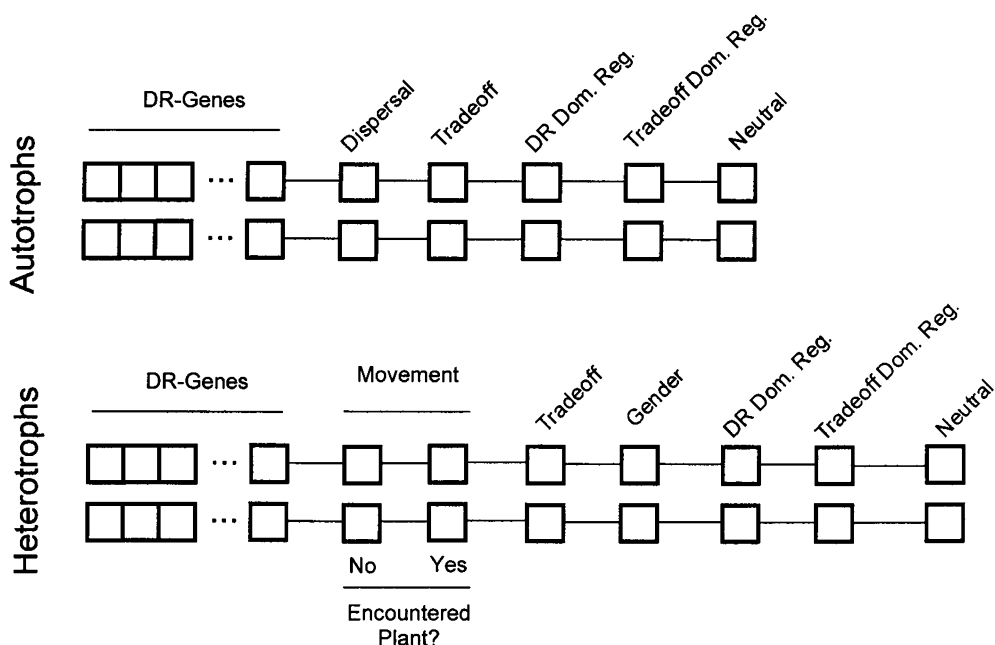


Figure 1. Chromosomal configuration for autotrophs and heterotrophs. Note that individuals are diploid and that loci are linked. Genes can be either "0" or "1". A "1" at a DR-genes locus codes for the expression of defense (autotrophs) or resistance (heterotrophs). Genes at the dominance loci code so that "00" would mean that "0s" are dominant, "11" would mean "1s" are dominant, and "01" and "10" genotypes cause the effected genes to function additively. Prior to mating a single crossing-over point is selected for recombination. A single chromosome is selected randomly for the gamete and joined to a similarly derived mate's chromosome. Mutations occurred with a probability of 0.001 at each locus converting a "0" to "1" or "1" to "0".

are useful in revealing the influence of spatial patterns on stochastic system dynamics, as are found in ecological and evolutionary systems [7]. Combining cellular automata with genetic algorithms began with Holland's [19, 20] Echo model. One advantage of lattice gas models, interacting particles systems, and Echo-type models over the pure cellular automata modeling technique is that they relax the dependence of simulated individuals on the grid.

Phenotypic interactions between parasites and their hosts have been modeled using single-locus, deterministic models [24, 26, 27]. However, an unexplored aspect is the response of increasing the number of loci involved in the coevolutionary interaction. Flor's [9] work on multilocus interactions suggests that multilocus models may be important in analyzing interdependent evolutionary systems. A recent trend in the literature has favored the investigation of multispecies interactions of diffuse coevolution [16, 32]. One of the goals of the model is to test the interplay between spatially dependent processes and the coevolutionary dynamics of plant-herbivore interactions.

3 The Model

Plants (autotrophs) and herbivores (heterotrophs) are represented explicitly by genomes that code for specific phenotypes (Figure 1). The individual organisms are diploid, with two homologous chromosomes. Two genes (alleles) can occur at each locus (either "0" or "1"). Genes that code for defense in plants and resistance in herbivores are collectively referred to as "DR-genes." A DR-gene of "1" codes for defense in autotrophs or resistance in heterotrophs. No defensive/resistant factor occurs from non-DR-genes

("0"). Autotroph chromosomes were composed of a linear arrangement of n -DR-genes that produced a defense against attacks by heterotrophs, and genes that coded for progeny dispersal distance from the mother, DR-tradeoff that determined the cost and benefit of producing a defensive product, two modifiers that determined dominance relationships (i.e., phenotype) of the DR-genes, and tradeoff genes that determined the cost of defensive genes. A nonfunctional gene was added to test neutral gene frequencies.

Heterotroph chromosomes also were composed of a linear arrangement of n -DR-genes that produced resistance to the defense produced by the DR-genes of the autotrophs. Heterotroph movement was determined by two genes: one expressed movement if the heterotroph encountered an autotroph in the last movement and the other was expressed if no autotroph had been encountered. Movement on the square lattice was straight ahead (01,10), spiral right (11), or spiral left (00). Analysis of variance was done using the mean tendency to spiral for each generation of heterotrophs, determined as

$$\text{Tendency to Spiral} = \left| 0.5 - \frac{\sum_{n=1}^N \left(\frac{S}{2} \right)}{N} \right|$$

where S is the number of genes coding for spiraling for each individual n in the population (N). The movement genes were followed by genes coding for resistance tradeoff, gender, two dominance modifier genes for DR-genes and tradeoff genes, and a neutral gene (Figure 1).

All genes for the initial founder populations were randomly assigned either 0 or 1. The functioning of genes coding for interaction between autotrophs and heterotrophs depended on the genes at the DR-dominance regulator locus. The explicit rules governing dominance and the costs of interaction were avoided by allowing these traits to evolve.

The organisms interacted on a spatially explicit, two-dimensional lattice that could be varied in size, shape (square or rectangular), and function (either a torus, where organisms reappeared on the opposite side of the lattice after contact with the boundary, or an island, where death resulted from a boundary breach). Resources for autotrophs were homogeneous and superabundant. Although autotrophs did not compete for resources, they did compete for progeny dispersal sites. Autotrophs were monoecious, having both sexes on each individual. Heterotrophs were dioecious. Both autotrophs and heterotrophs were obligate out-crossers. A single generation for autotrophs and heterotrophs involved seven movements by heterotrophs, with fitness scores tallied after each movement time step. Seven movement steps allowed spiraling heterotrophs to explore their local neighborhood without encountering previously visited cells. During each movement step interactions between autotrophs and heterotrophs were analyzed and fitness scores were adjusted. Figure 2 presents pseudocode depicting the model's structure.

The model allowed for the number of autotroph and heterotroph generations to be varied independently. To test the effect of increased herbivore-to-plant generation times we maintained a single autotroph generation per period and tested five different heterotroph generations per period (Table 1). Therefore, for these tests, autotroph generation and period are synonymous. Generations for both autotrophs and heterotrophs were discrete. Simulations involved 1,200 periods, although the analyses were done using only the last 600 periods unless otherwise noted (Table 1). Four replicate sim-

```

create individuals on grid
for i = 1 to # periods
{
  for j = 1 to # heterotroph generations per period
  {
    for k = 1 to # heterotroph movements per generation
    {
      determine fitness of autotrophs and heterotrophs
      calculate statistics
      move heterotrophs
    }
    pair heterotrophs and mate (recombination and mutation)
    place new heterotrophs on grid
  }
  pair autotrophs and mate (recombination and mutation)
  place new autotrophs on grid
}

```

Figure 2. Pseudocode depicting the flow of the model. We also were able to have multiple autotroph generations per period but these tests were not done and, therefore, are not represented in code.

Table 1. Model settings for the test conditions, including grid size, shape, and function, and the number of replicate simulations performed. "H" and "A" refer to heterotrophs and autotrophs, respectively. For habitat shape and function tests organisms had eight DR-loci.

Test	Grid	Test Details	Number of Periods	Number of Replicates
Control	40 × 40	No Heterotrophs	1200 ^a	6
Habitat Shape and Function	50 × 50	Torus	2000	4
	50 × 50	Island	2000	4
	100 × 25	Torus	2000	4
	100 × 25	Island	2000	4
On a Torus Lattice:				
Number of Interaction Loci	40 × 40	1,2,4,8,16 loci	1200 ^a	6
Heterotroph:Autotroph Generations	40 × 40	1:1,4:1,16:1,64:1	1200 ^a	6
Period ⁻¹		250:1	400 ^b	6

^a Analyses were done using the last 600 periods.

^b We ran these simulations for 400 periods because each simulation required 16 hours on a Sun Sparc20 with 160 megs of RAM. Analyses were done using the last 200 periods.

ulations were performed for each of the habitat shape/function tests. Six replicate simulations were performed for controls and for each test of the various number of DR-loci and the tests of the ratios of heterotroph:autotroph generations. One set of experiments testing the coevolutionary interaction when heterotrophs had 250 generations to each autotroph generation was run for 400 periods (autotroph generations) instead of 1,200 periods due to the computer time required for each replicate simulation (16 hours/replicate on a Sun SPARC20 with 160 Megs RAM) (analyses done on the last 200 generations). These latter simulations produced about 100 million heterotrophs over 100,000 generations. The program was written using the Pascal programming language.

3.1 Selection and Mating

Fitness score increases for heterotrophs were accumulated by encountering autotrophs (occupying the same grid cell). The benefit from each encounter depended on autotroph defense, and the heterotroph's ability to resist autotroph defenses. It is important to note that costs and benefits were determined for each individual and depended on genes at the DR-loci, the DR-dominance regulator locus, and the tradeoff locus. During encounters autotroph and heterotroph genomes were compared locus against locus (gene-for-gene) rather than comparing whole-organism phenotypes. In this way, autotrophs and heterotrophs were considered as having multiple independent phenotypes (e.g., multiple compounds in autotrophs and heterotrophs).

If, for example, an individual heterotroph carried no genes for detoxification at a particular locus (00), it would not accumulate detoxification enzyme costs for that locus. If the encountered autotroph was undefended (00) the autotroph would have no fitness increment while the heterotroph would gain a relatively high fitness score (Figure 3). Such heterotrophs (00) would gain little, however, from an encounter with well-defended autotrophs (e.g., 11, or 10/01 with dominant defensive genes) (Figure 3). Cost for interactions depicted in Figure 3 are considered additive but were actually susceptible to changes in dominance, regulated by the DR-dominance modifier genes. Figure 4 shows a hypothetical encounter between an autotroph and a heterotroph, the resulting fitness scores for each of the eight loci, and the total fitness increments for these individuals. Individual fitness scores, therefore, depended on the sum of fitness gains over all DR-genes (evaluated gene-for-gene) with all encountered herbivores (for plants) or plants (for herbivores). Scores for autotrophs that escaped herbivory were considerably higher with slight reductions due to the cost of defense.

A mixed hard and soft selection scheme was used to determine which individuals would reproduce (see [12]). Individual autotrophs and heterotrophs lacking a minimum fitness score did not reproduce (hard selection). The remaining individuals were ranked according to their fitness score (soft selection). Autotrophs and heterotrophs with higher fitness scores at the end of each generation had a higher probability of being chosen to participate in matings but did not necessarily have higher realized fitness (actual number of progeny produced) due to density-dependent space limitations for progeny dispersal and the probability of simply not being selected to mate by chance. After randomly choosing an individual for reproduction according to fitness score (specifically a female for heterotrophs), a mate (likewise, a male for heterotrophs) was chosen by searching neighboring cells, spiraling outward without exceeding a distance of three cells away (i.e., not more than the closest 48 cells). The total number of progeny produced was determined using the logistic growth equation. Autotroph and heterotroph carrying capacities were set to 320 and 960 (20% and 60% of the lattice), respectively. The higher carrying capacity for heterotrophs ensured encounters between autotrophs and heterotrophs. High intrinsic growth rates were used for both

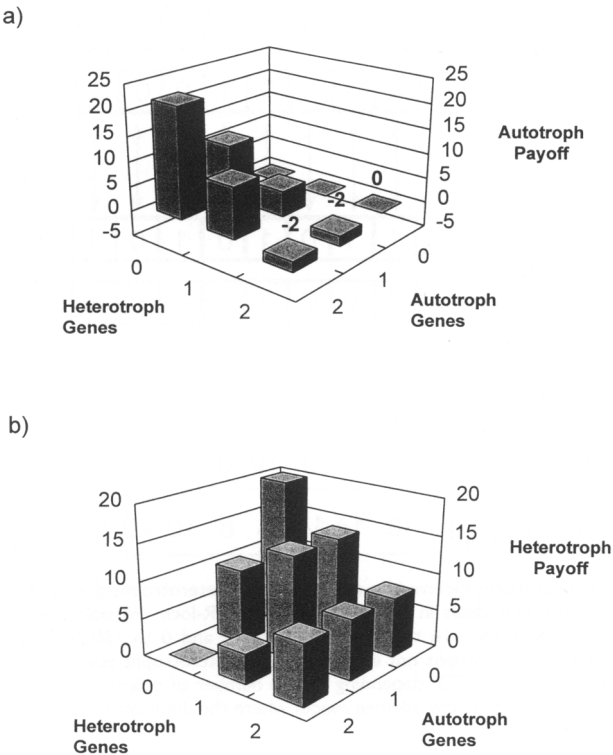


Figure 3. Payoffs during an encounter for both autotrophs and heterotrophs assuming the default tradeoff values (see Table 2). Values on the z-axis represent fitness score increments (or decrements) per locus for each individual. The calculation of a hypothetical encounter is depicted in Figure 4. Relative fitness scores for encounters follows a strictly additive model, which is not the case when the dominance regulator genotype is either 00 or 11. Without an encounter herbivores accumulate no fitness score and autotrophs accumulate scores based on their individual levels of defense and whether genes are dominant for either expression (1) or no expression (0) of defensive genes.

autotrophs and heterotrophs, making populations for both species rather stable at or near their respective carrying capacities.

Individuals chosen for mating underwent meiosis and offspring were formed by randomly combining one recombinant chromosome from each parent. Recombination included a single crossover event at a random site on the chromosome and random mutation, with a probability of 0.001 for each locus. Mutation was achieved by switching genes from “0” to “1” or vice versa. Autotroph progeny were placed at a distance determined by the mother’s dispersal chromosome (i.e., 00, 01, 10, and 11 coded for 1, 2, 2, and 3 cells away, respectively). Direction for seed dissemination was a randomly chosen diagonal and, if the cell was occupied, the next available cell was chosen in a clockwise direction. Heterotroph progeny were placed around the female, with successive progeny placed in available cells radiating outward from the mother. If an open cell was not found within a 4-cell radius (81 possible cells), no more progeny were produced by that female.

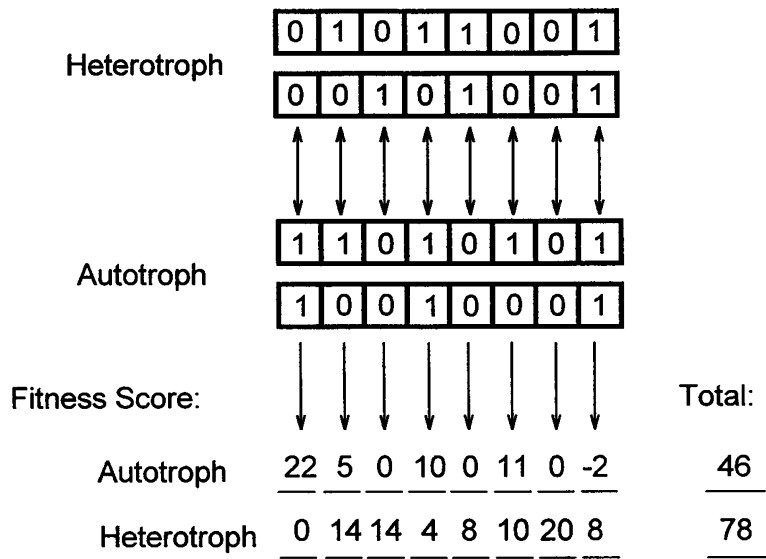


Figure 4. Fitness scores received for autotrophs and heterotrophs during a hypothetical encounter. The individuals in this test have eight DR-loci. Possible total scores for encounters range from -16 to 176 for autotrophs and 0 to 160 for heterotrophs. Fitness scores for autotrophs not encountered by heterotrophs (example not shown) range between 256 and 272, depending on the number of defensive genes expressed. These examples use a strictly additive model where the dominance regulator genotype is either 01 or 10.

3.2 Time-Series Analysis to Determine Coevolutionary Interaction

Coevolution between autotrophs and heterotrophs was tested using cross correlation of the time series between the number of DR-genes for autotrophs and heterotrophs. Initial analysis of the number of DR-genes for both autotrophs and heterotrophs from generation 600 to 1,200 showed them to behave as a stationary process. We therefore used the CROSSCOR option of the procedure ARIMA of the statistics package SAS without differencing to find lag periods with significant correlations. The lag period with the greatest correlation ($r = 0.48$ to 0.83) was recorded for each replication for the ratio simulations. Analysis of variance was used to test for differences in mean lag times between simulations varying in the ratio of heterotroph to autotroph generations per period. For this last analysis we tested three randomly chosen simulations from the ratio tests 1:1, 4:1, 16:1, and 64:1 heterotroph to autotroph generations per period. A second time series analysis was conducted on the correlation between the mean number of neutral genes of autotrophs and heterotrophs in order to contrast with the corresponding analysis on DR-genes.

4 Testing the Model

We tested the following five questions regarding the interaction between autotrophs and heterotrophs:

- 1. Does the number of genes involved in autotroph defense and heterotroph resistance affect the coevolutionary response between autotrophs and heterotrophs? We tested the coevolutionary response of 1, 2, 4, 8, and 16 loci involved in this interaction.

2. How does the ratio of the number of heterotroph:autotroph generations affect the evolution of defense and resistance in autotrophs and heterotrophs, respectively? We tested this relationship by successively increasing the number of heterotroph generations per period, leaving autotrophs to reproduce once each period (i.e., 1:1, 4:1, 16:1, 64:1, and 250:1 heterotroph:autotroph generations).
3. What is the effect of habitat shape (square or rectangular) and function (torus or island) on the stability of the plant-herbivore system? The grid was changed from a square (50×50) to rectangle (100×25) without altering the total number of cells. We also tested the stability of the interaction when the habitat functioned as an island (boundary breach resulted in death) and a torus (heterotrophs or plant progeny would appear on the other side of lattice upon boundary breach).
4. How does the encounter rate between heterotrophs and autotrophs affect autotroph progeny dispersal distances? Genes coding for autotroph progeny dispersal distances were tested for response to changing encounter rates.
5. Finally, how does this coevolutionary system behave qualitatively over thousands of generations? This question was evaluated by observing mean number of DR-genes between autotrophs and heterotrophs.

5 Results and Discussion

Autotroph fitness and the production of defensive compounds (number of defensive genes) were highly sensitive to low levels of heterotroph activity. After initial adaptation to relatively low encounters between species the mean number of DR-genes in both autotrophs and heterotrophs was dynamic over time but the overall trend remained constant (statistically stationary). Without heterotrophs, mean plant fitness was relatively high (Figure 5) with strong selection against genes coding for defense (Figure 6). With only a single DR-locus, defensive genes apparently were not adaptive (Figure 6). In replicate simulations with a single locus DR-genes consistently were lost, indicating selective pressure rather than drift was responsible for their elimination. Had drift been important we would have expected high variance in the number of genes expressed. Apparently two or more loci are required to maintain defensive genes over hundreds of generations. These data suggest that in environments where herbivores negatively affect plant fitness selection should increase the number and diversity of plant defenses. In addition, these results suggest that plants are likely to be highly sensitive to even low levels of herbivory and would be expected to maintain a relatively diverse array of defenses. At a critical low level of herbivory, however, selective pressures might effectively eliminate defensive genes from the plant population, although this response is highly dependent on the cost of defense. In our simulations the cost of defense for autotrophs was set at a very low level relative to costs of resistance in heterotrophs (Table 2). This situation likely decreased the model's sensitivity to detecting selective pressure on the maintenance of defensive genes in autotrophs (defense against herbivores was cheap), thus suggesting that these results are robust.

5.1 The Effect of Varying the Ratio of Heterotroph:Autotroph Generations

The encounter rate between autotrophs and heterotrophs increased with the increasing ratio of the number of heterotroph:autotroph generations (Figure 5). Over successive generations heterotrophs concentrated and reproduced more successfully in areas of the grid dominated by autotrophs. In concert with heterotrophs congregating near autotrophs, the genes coding for heterotroph movement indicated a selective advantage for individuals to spiral both after encountering an autotroph and after failing to

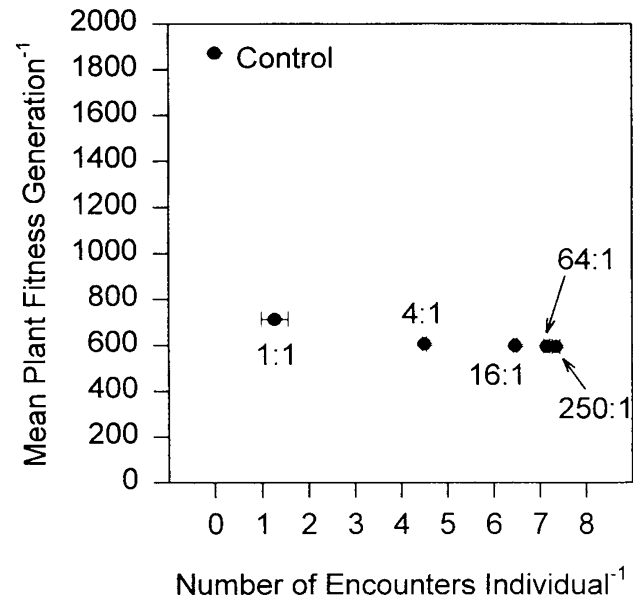


Figure 5. Mean plant fitness score plotted against the mean number of autotroph encounters by heterotrophs for different ratios of heterotroph:autotroph generations. "Control" indicates simulations where heterotrophs were not present on the lattice. Error bars are ± 1 standard deviation. $N = 6$ for each point.

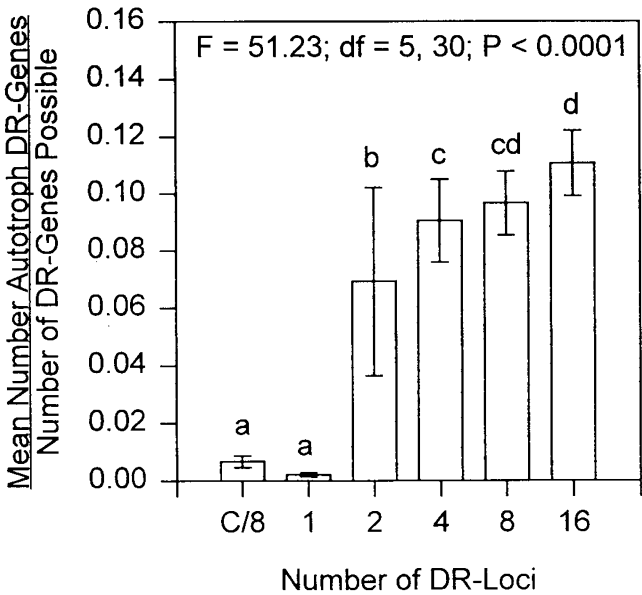


Figure 6. The number of autotroph DR-loci as a proportion of possible genes plotted as a function of the number of DR-loci. Relatively few DR-genes are present with only a single locus. "C/8" represents simulations where heterotrophs were not present and autotrophs had eight defensive loci. Means with different letters are statistically different based on least significant differences. Error bars are ± 1 standard deviation. $N = 6$ for each simulation.

Table 2. The values of parameters assigned for the genetic model determining fitness for autotrophs (A) and heterotrophs (H).

Genetic Model Parameters	Value
Cost of A Defense	1
Cost of H Resistance	6
A and H Tradeoff ^a	2
A Benefit Despite Encounter	24
A Benefit for No Encounter	34
H Benefit for Encounter	20
H Benefit for No Encounter	0

^a Tradeoff is determined by genes at a single locus for each individual and controls the magnitude of the inverse relationship between cost and benefit of carrying DR-genes.

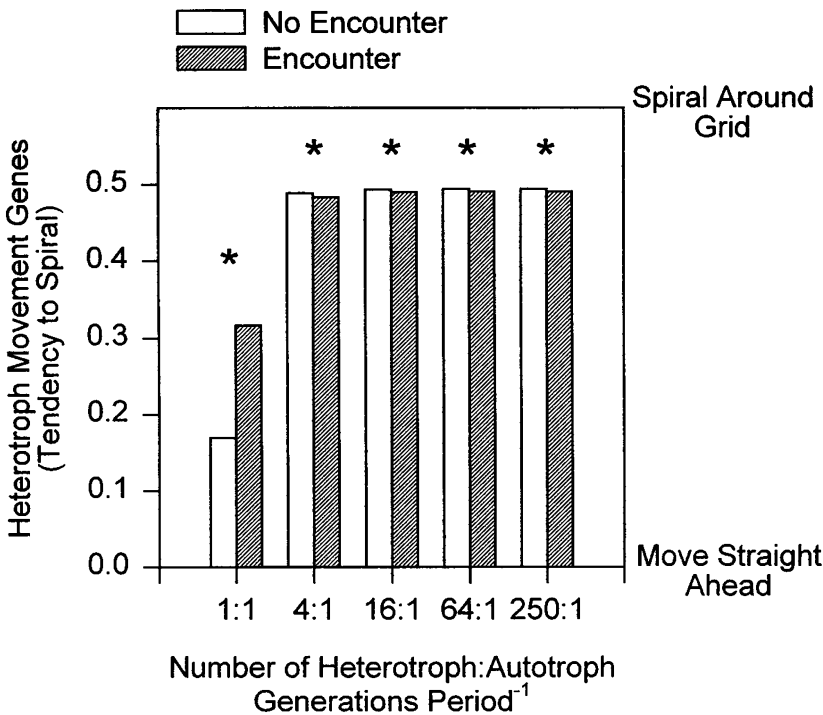


Figure 7. The tendency for heterotrophs to spiral either left or right that either did or did not encounter an autotroph in the last movement. The “*” indicates within generation ratio differences between means using least significant differences.

encounter an autotroph (Figure 7). The mean tendency to spiral for heterotrophs increased significantly when the heterotroph to autotroph ratio of generations increased from 1:1 to 4:1 ($P < 0.001$). In addition, at relatively low encounter rates (ratio of 1:1) heterotrophs spiraled more when autotrophs were encountered. At higher ratios (4:1 through 250:1) heterotrophs spiraled more if they did not encounter autotrophs ($P < 0.001$). The importance of the results for ratios greater than 1:1 appears to be statistical rather than biological. The significant interaction between ratio and whether

Table 3. Analysis of variance for the mean tendency to spiral in heterotrophs after encountering or not encountering an autotroph. The factor "ratio" refers to the number of heterotroph to autotroph generations per period (i.e., 1:1, 4:1, 16:1, 64:1, and 250:1) (see Figure 7).

Source	DF	Sums of Squares	F	P
Ratio	4	0.584	96.34	< 0.0001
Encounter	1	0.011	7.26	< 0.01
Ratio $\times$ Encounter	4	0.054	8.95	< 0.0001
Error	48	0.073		

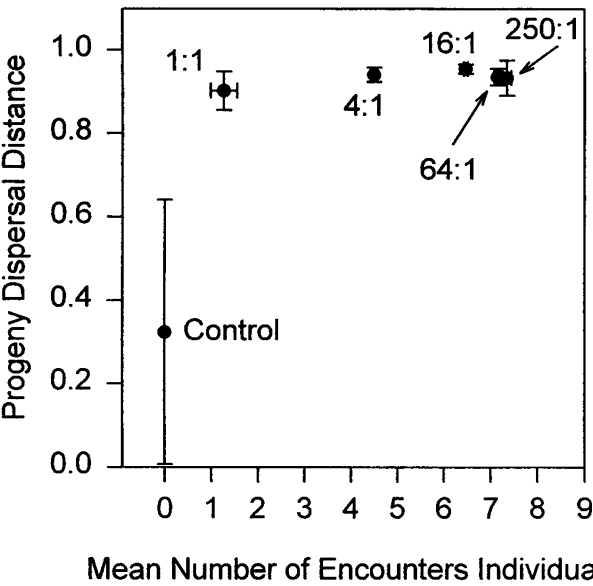


Figure 8. Progeny dispersal distance as a function of autotroph encounters by heterotrophs for the increasing ratio of heterotroph:autotroph generations. Controls lacked heterotroph populations. Error bars are ± 1 standard deviation.

the heterotroph encountered an autotroph (Table 3) suggests that the movement of heterotrophs is dependent both on their rate of encounters with host plants and the relative rates of reproduction (number of heterotroph to autotroph generations per period). It is also likely that the relationship of the relative rates of reproduction between plants and herbivores affects the genetic interaction of defense and resistance [22], and these data support this suggestion.

Herbivory increased autotroph progeny dispersal distance (Figure 8). In absence of heterotrophs, autotroph progeny dispersal distance was significantly lower than at higher rates of herbivory. At the higher encounter rates (64:1 and 250:1 heterotroph:autotroph generations per period), however, there was a decline in autotroph dispersal distance (Figure 8). Maximum dispersal distances at intermediate encounter rates suggests a tradeoff between colonization (distance progeny are dispersed) and competitive ability (plant fitness) [34]. The main trend of the inverse relationship between dispersal distance and plant fitness suggests that the colonization/competition tradeoff in this model is coupled to heterotroph activity. If plants in areas of high herbivory reduce defense and maximize growth, we may expect to see either an increase in competitive ability for space or an increase in colonization ability and dispersal

distance. We hypothesized that autotroph progeny dispersal distance would increase under increasing heterotroph attack but did not expect a decline at the highest tested encounter rate. These data suggest that dispersal distance is coupled to the intensity of herbivory with plants under severe herbivory dispersing progeny somewhat closer to the parent.

Plants, obviously, cannot explore sites in ecological time as do many animals. Plant species, however, do search the fitness landscape for optimal dispersal distance in evolutionary time. In a comparative study Gleeson and Tilman [14] found a tradeoff between colonization rate and competitive ability in a grassland in Minnesota. Their findings suggest that species with high colonization rates will be superior in disturbed habitats, a condition that may be caused by intense herbivory. Plant species that are competitively superior for limiting resources often are more persistent when disturbance rates are low but may do poorly when disturbed by herbivores [17]. Therefore, as this model suggests, herbivore populations may evolve the ability to resist plant defenses and, therefore, cause increased damage and disturbance while plants evolve increased dispersal distances. These results support Gleeson and Tilman's model. Our model, however, separates colonization into the rate of propagule production and seed dispersal distance. The relationship between dispersal and competitive ability is further complicated by plant longevity, which was not tested in this model [34].

We found that plant fitness declined as the ratio of heterotroph:autotroph generations increased. The greatest difference in plant fitness, however, was between controls lacking heterotrophs and when heterotroph and autotroph generations were equally frequent (Figure 5). The structure of the fitness landscapes for both heterotrophs and autotrophs are highly dependent on the spatial and genetical relationships between organisms within species and between trophic levels [23, 28]. The frequency of defensive genes in autotrophs in absence of heterotrophs was maintained at very low levels, most likely due to mutation (Figure 6). The most parsimonious explanation for this is that selection reduced the occurrence of genes coding for resistance due to the associated physiological cost of defense. The assumption that plant defenses are costly, however, may not be universally appropriate. Baldwin et al. [3], for example, found that varying alkaloid concentrations in tobacco (*Nicotiana glauca*) does not affect rates of carbon gain, although he did find an associated fitness cost with nicotine production. He suggested that the primary function of nicotine may be to store nitrogen, an essential plant nutrient, rather than to defend against herbivory. Our results, which depend on this cost assumption, indicate that the evolution of plant defense to herbivory is more sensitive to the presence of herbivores than to the number of generations herbivores have for each plant generation.

5.2 The Number of DR-Gene Loci Affects Plant-Herbivore Coevolution

There was a positive relationship between the proportion of defensive or resistance genes and the number of DR-loci (Figure 6). With only one DR-locus autotrophs and heterotrophs had relatively few DR-genes. The explanation for this does not appear to be loss of genes due to drift because replicate simulations behaved similarly and background mutation rates continued to provide adequate variation for selection to favor DR-genes. When the number of loci exceeded one, DR-genes had higher relative frequencies (Figure 6). This finding suggests that the evolution of gene-for-gene interactions requires a sudden saltation of two or more loci in order to generate sufficiently large numbers of DR-genes and their products. The development of a single locus does not seem adequate to provide defense against herbivore attack. We, therefore, would predict multiple genes to dominate gene-for-gene systems, as is the case [33].

In addition, the increased number of loci, which also resulted in an increase in the relative number of DR-genes, was inversely related to the relative amounts of variation

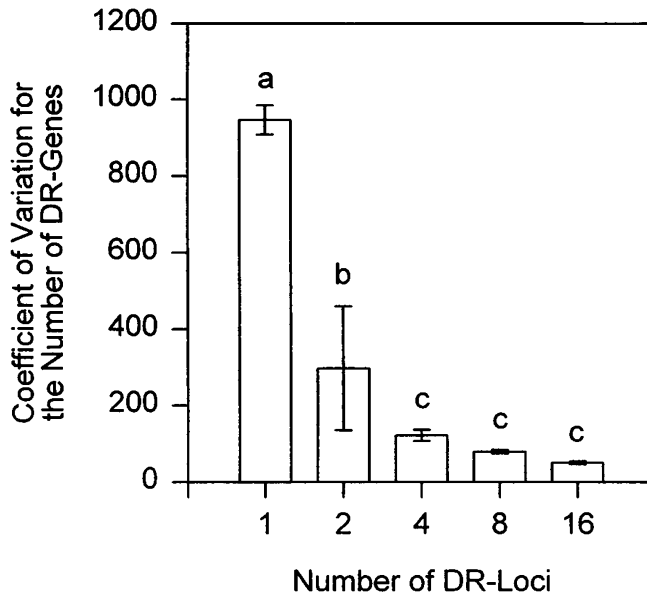


Figure 9. The coefficient of variation (relative to the mean number of autotroph DR-genes) as a function of the number of DR-loci. With the increasing number of loci the variation from generation to generation in autotrophs tends to decline. Means with different letters are statistically different based on least significant differences. Error bars are ± 1 standard deviation.

Table 4. Analysis of variance for the mean number of periods simulation ran (out of a possible 2,000). Heterotrophs and autotrophs each had one generation per period (see Figure 10).

Source	DF	Sums of Squares	F	P
Shape	1	1282556	15.71	< 0.01
Function	1	4654806	57.02	< 0.0001
Shape $\times$ Function	1	1282556	15.71	< 0.01
Error	12	979575		

among autotrophs in the number of genes expressed (Figure 9). This suggests that the autotroph population was more able to achieve an optimal level of defense when they had more loci for defensive genes, interpreted from reduced variation of the number of genes within populations.

5.3 Habitat Shape and Function Affected Plant-Herbivore Coevolution

Community stability, measured as the number of generations of autotroph and heterotroph coexistence was sensitive to both habitat shape and function (Figure 10). We found a significant interaction between habitat shape (square or rectangular) and function (island or torus) (Table 4). Populations were most stable when the habitat functioned as a torus. The average number of generations until a species (always the heterotroph) became extinct decreased when the habitat functioned as an island. The island habitat was significantly more stable when square (50 $\times$ 50) than rectangular

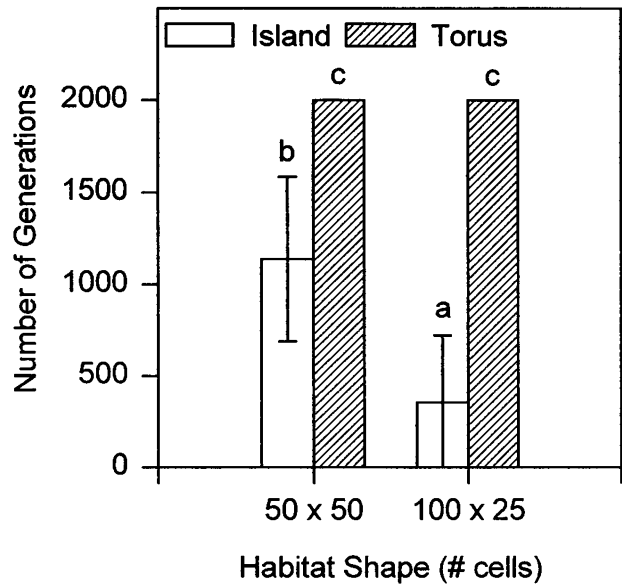


Figure 10. The mean number of generations that simulations continued without extinction. Simulations should have continued for 2,000 generations. Means with different letters are statistically different based on least significant differences. Error bars are ± 1 standard deviation.

(100 $\times$ 25) (Figure 10). These results are consistent with current biogeography theory that suggests higher turnover and greater instability of populations on islands than on large, mainland habitats [4]. The ability of coevolutionary responses to offset instabilities in species diversity, however, may be of limited importance in disturbed habitats. This model suggests that the influence of interacting habitat shape and function and plant-herbivore coevolution on community stability is important in maintaining species diversity. In a test of model food webs, Freeland and Boulton [10] found that webs that had coevolving species were more stable than webs lacking coevolutionary responses. Taken together, the stability of ecological communities is likely to be dependent both on the shape and function of the habitat and the level of coevolutionary coupling between interacting species. In addition, the effect of species' coevolutionary interactions in real environments on the community stability is currently unknown. The possibility that disturbances may cause catastrophic cascading effects through trophic food webs may be increased or decreased by coevolutionary connectedness between species. This important relationship remains to be tested.

For habitats (long and thin or square) that functioned as islands, heterotroph populations maintained relatively high levels of defensive genes while autotrophs were relatively undefended. There appears to be a selective advantage for herbivores to carry more resistance genes in island habitats relative to larger habitats. In addition, defensive genes are lost from the autotroph populations more quickly than heterotrophs, irrespective of lattice function. In general, encounter rate between heterotrophs and autotrophs was reduced on island habitats so that the cost of defense outweighs the benefit of protection against damage by heterotrophs.

5.4 Evidence for Coevolution

Cross-correlation analysis revealed a significant lag between heterotroph and autotroph mean number of DR-genes, with heterotrophs trailing autotrophs. The lag between

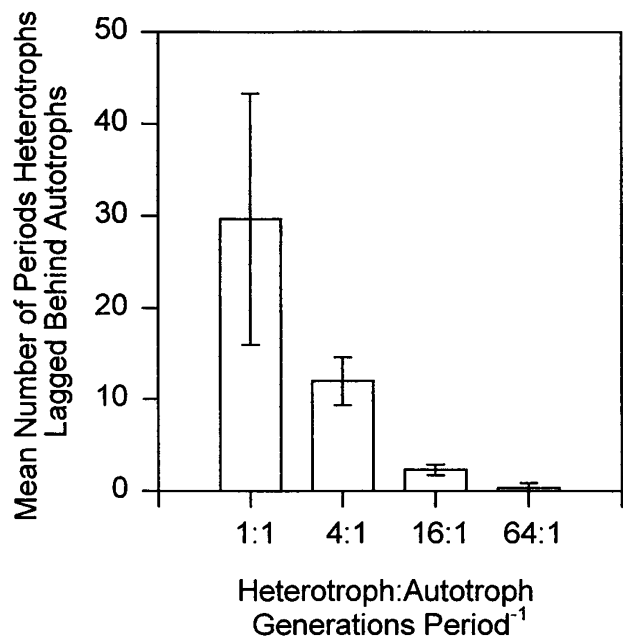


Figure 11. The number of periods that the DR-gene frequency response of heterotrophs lagged behind autotrophs.

Table 5. Analysis of variance for the mean number of periods that the heterotrophs lagged behind autotrophs in the number of DR-genes expressed. The lag significantly decreased with increased heterotroph:autotroph generations per period (see Figure 11).

Source	DF	Sums of Squares	F	P
Ratio	3	1614.92	11.10	< 0.01
Error	8	388.00		

species decreased with increasing ratio of heterotroph to autotroph generations per period (Figure 11; Table 5). We found no significant trend for any lag periods in the test on coevolution between neutral genes and, statistically, the neutral gene time series behaved as white noise.

There was a positive relationship between the mean number of DR-genes expressed by autotrophs and heterotrophs for the last 50 generations for each of the simulations varying the number of DR-loci (Figure 12). The increasing trend, plotted with a second-order polynomial, suggests that with increasing numbers of loci involved in the gene-for-gene interaction, heterotrophs increase expression of resistance genes relative to defensive genes in autotrophs. The significance of this relationship may lead to selection in autotrophs to maximize the number of defensive loci because the cost for resistance, at least in this model, increased more rapidly for heterotrophs than the cost for defense in autotrophs.

These analyses demonstrate four points. First, there was coevolution for DR-genes between autotrophs and heterotrophs, seen in the significant cross correlations. Second, heterotrophs produced more resistance relative to autotroph defense with increasing numbers of loci in the gene-for-gene interaction. This suggests a selective advantage for plants to maximize the diversity of defenses against herbivores. Third, the lag with the

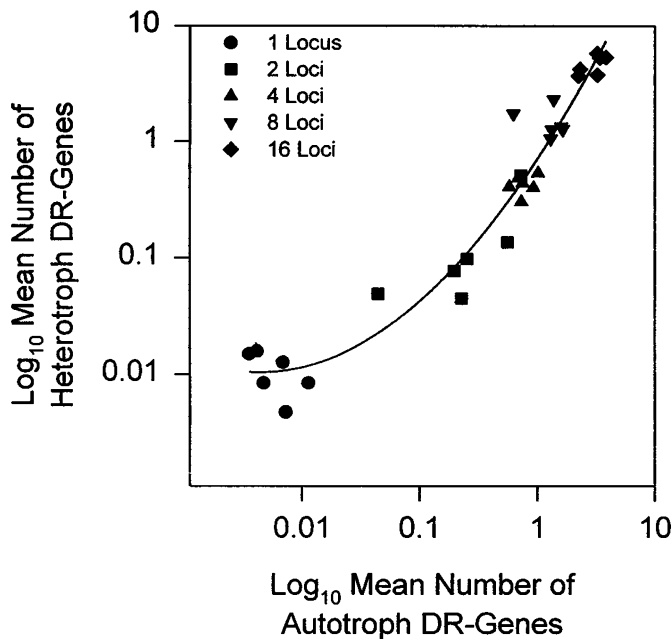


Figure 12. Relationship between the $\log_{10}$ mean number of heterotroph resistance genes plotted against the $\log_{10}$ mean number of autotroph defense genes. The means were determined from the last 50 of 1,200 generations for all replicates from the DR-loci tests. The positive relationship ($R^2 = 0.95$) indicates coevolutionary coupling of autotrophs and heterotrophs.

greatest correlation was longer when the ratio of heterotroph to autotroph generations per period was similar. Stated in another way, as the ratio of heterotroph to autotroph generations increases the coevolutionary response between species decreases in time. Finally, neutral genes do not coevolve and illustrate the expected response of genes not under selection.

5.5 Plant-Herbivore Coevolution Was Dynamic Over Thousands of Generations

DR-gene frequencies for heterotrophs and autotrophs fluctuated throughout all simulations, and are represented by the dynamic response of heterotrophs and autotrophs where the ratio of heterotroph to autotroph generations was 250:1, resulting in 100,000 heterotroph generations (Figure 13). This surprising result suggests that we should not expect coevolution between plants and herbivores for defense and resistance to reach a stable equilibrium, even when herbivores have hundreds of generations per host generation. Through inspection of the lattice, generation to generation, it was possible to discern the spatial structure of the interacting populations, where pockets of autotrophs and heterotrophs assembled and interacted. Because DR-genes in autotrophs in absence of heterotrophs are maintained at very low levels due to the interaction of mutation and negative selection due to cost and recognizing the fact that the lattice was homogenous, we may conclude that the maintenance of DR-gene fluctuations resulted from the interaction of space (individuals occupying specific locations on the lattice with local effects) and time (the ability of plants and herbivores to evolve).

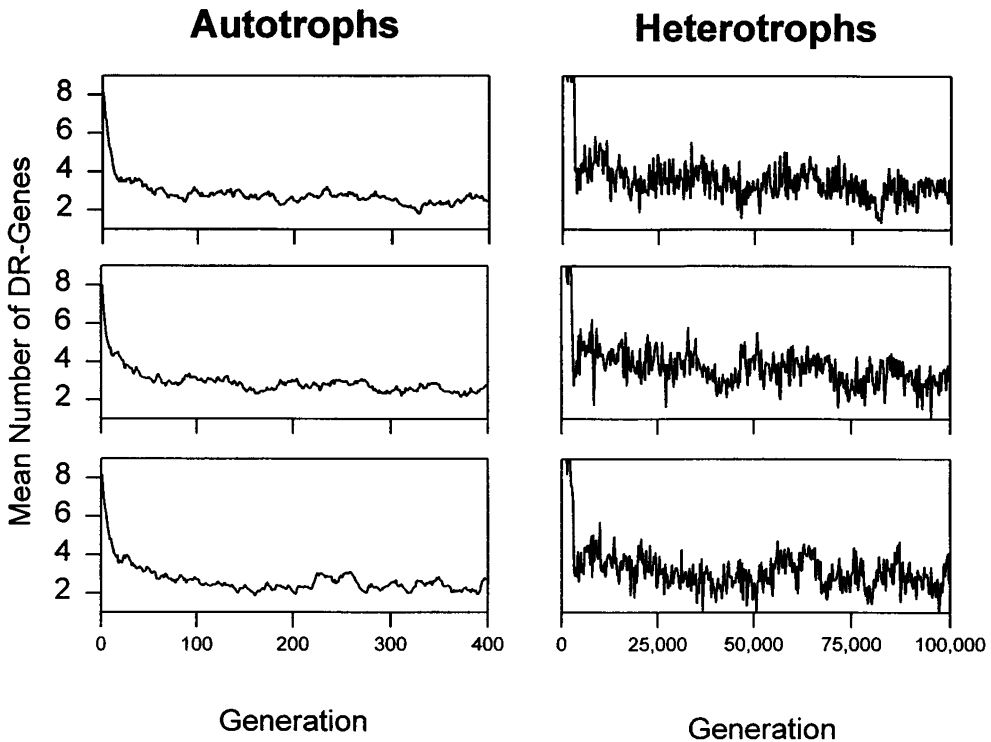


Figure 13. Three paired replicate simulations for the mean number of DR-genes in autotrophs and heterotrophs when the ratio of heterotroph to autotroph generations per period was 250:1.

6 Conclusions

We present a spatially explicit coevolutionary model testing the effects of changing the number of loci for genes coding for defense in plants and resistance in herbivores, changing the ratio of the rates of reproduction between plants and herbivores, and altering habitat structure and function on the genetic interaction between plants and herbivores. The results of this model suggest that, on a simple, relatively small homogenous grid, plant-herbivore coevolutionary interactions are complex and dependent on habitat shape and function. Most interesting, perhaps, is that the model suggests plant-herbivore coevolution is very sensitive to low levels of interaction. Increasing the number of loci involved in the gene-for-gene coevolutionary interaction from one to two was most important in affecting organism fitness and the evolution of plant defense and herbivore resistance. Increasing the ratio of herbivore to plant generations per period from 1:1 to 4:1 resulted in a substantial increase in the coupling of coevolutionary responses (decreased time lag between herbivore response to changes in plant defense). These findings suggest an important interplay between the spatial relationships among interacting organisms and the genetic makeup of the coevolving species (e.g., the number of loci involved in gene-for-gene coevolution).

Previous theoretical work has suggested that coevolving species may be more likely to go extinct in response to reductions in habitat size [18]. This finding is consistent with current biogeographic theory but does not address the influence of changes in habitat shape and function. This paper demonstrates that stability, measured as the reciprocal of extinction probability, is reduced by increasing the ratio of edge to area

and the level of habitat isolation. Finally, the results from this model suggest that the ability of interacting species to evolve in response to various perturbations is important and should be included in community-level models that test long-term dynamics to various factors. For instance, there currently exists no theoretical framework in which to consider the effects of increasing atmospheric CO₂ on the stability of communities comprising coevolving species. In addition, large-scale perturbations are likely to affect food-web dynamics, such as increasing or decreasing the likelihood of catastrophic cascades. The approach taken in this model could be modified to test these questions.

Acknowledgment

We would like to thank Academic Computing Services and Faculty Computing Services of Syracuse University for help and use of their computer equipment. The manuscript was greatly improved by the helpful comments of Rick Wahle and two anonymous reviewers. Partial funding for this project was received by WTS from NSF.

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